

Bash scripting -2

1. Create a bash script to check if a directory is available or not.

```
[root@ip-172-31-35-143 scripts]# mkdir /abhishek/test
[root@ip-172-31-35-143 scripts]#
```

```
#!/bin/bash

DIR="/abhishek/test"

if [ -d "$dir" ]
then
echo "Directory exists."
else
echo "Directory does not exist."
fi
```

Create a abhishek.bash give 755 permissions Check the output using ./abhishek.bash

```
[root@ip-172-31-35-143 scripts]# vi abhishek.bash
[root@ip-172-31-35-143 scripts]# chod 755 abhishek.bash
bash: chod: command not found
[root@ip-172-31-35-143 scripts]# chmod 755 abhishek.bash
[root@ip-172-31-35-143 scripts]# ll
total 8
-rwxr-xr-x. 1 root root  51 Sep  8 04:40 abhishek.bash
-rw-r--r--. 1 root root 128 Sep  8 04:39 directory.txt
[root@ip-172-31-35-143 scripts]# ./abhishek.bash
Hello, this is  abhishek.bash
[root@ip-172-31-35-143 scripts]# vi abhishek.bash
[root@ip-172-31-35-143 scripts]# cat abhishek.bash
#!/bin/bash

echo "Directory exists"
[root@ip-172-31-35-143 scripts]# ./abhishek.bash
Directory exists
```

2. Create a bash script to create multiple files

```
#!/bin/bash

touch  abhi1.txt abhi2.txt abhi3.txt

echo " Multiple files created succesfully "
```

```
echo " Multiple files created succesfully "
```

3. Create a bash script to take a backup of a directory.

```
#!/bin/bash
tar -czf backup.tar.gz /abhi/test
echo "Backup successfully completed"
```

```
[root@ip-172-31-35-143 ~]# vi backup.bash
[root@ip-172-31-35-143 ~]# vi backup.bash
[root@ip-172-31-35-143 ~]# chmod 755 backup.bash
[root@ip-172-31-35-143 ~]# ./backup.bash
tar: Removing leading `/' from member names
tar: /abhi/test: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
Backup successfully completed
[root@ip-172-31-35-143 ~]# ls -lh backup.tar.gz
-rw-r--r--. 1 root root 45 Sep  8 05:48 backup.tar.gz
[root@ip-172-31-35-143 ~]# |
```

4. Create a bash script to install Nginx on an EC2 server

```
#!/bin/bash
sudo dnf install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
echo " nginx installed successfully"
```

```
[root@ip-172-31-35-143 ~]# vi backup.bash
[root@ip-172-31-35-143 ~]# vi nginx.bash
[root@ip-172-31-35-143 ~]# chmod 755 nginx.bash
[root@ip-172-31-35-143 ~]# ./nginx.bash
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
492 kB/s | 69 kB | 00:00
-----
Package                               Architecture      Version           Repository        Size
-----
Installing:
nginx                                  x86_64            1:1.30.4-1.amzn2023.0.1  amazonlinux      34 k
Installing dependencies:
generic-httpd                         noarch            18.0.0-12.amzn2023.0.3  amazonlinux      19 k
perltools-libos                       x86_64            2.9.1-1.amzn2023.0.3    amazonlinux      308 k
libunwind                             x86_64            1.4.0-5.amzn2023.0.3    amazonlinux      66 k
nginx-core                            x86_64            1:1.30.4-1.amzn2023.0.1  amazonlinux      711 k
nginxfilesystem                      noarch            1:1.30.4-1.amzn2023.0.1  amazonlinux      10 k
nginx-mimetypes                      noarch            2.1.49-3.amzn2023.0.3    amazonlinux      21 k
Transaction Summary
-----
Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
-----
```

5. Create a bash script to install Apache Tomcat on an EC2 server.

```
#!/bin/bash

dnf install java-17-amazon-corretto -y

dnf install tomcat9 -y

systemctl start tomcat9

systemctl enable tomcat9
```

```
[root@ip-172-31-32-133 ~]# vi tomcat.bash
[root@ip-172-31-32-133 ~]# vi tomcat.bash
[root@ip-172-31-32-133 ~]# chmod 755 tomcat.bash
[root@ip-172-31-32-133 ~]# ./tomcat.bash
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
java-17-amazon-corretto	x86_64	1:17.0.20+10-1.amzn2023.1	amazonlinux	220 k
Installing dependencies:				
alsa-lib	x86_64	1:2.7.2-1.amzn2023.0.4	amazonlinux	503 k
cairo	x86_64	1.18.0-4.amzn2023.0.3	amazonlinux	717 k
dejavu-sans-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.3 M
dejavu-sans-mono-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	467 k
dejavu-serif-fonts	noarch	2.37-16.amzn2023.0.2	amazonlinux	1.0 M
fontconfig	x86_64	2.13.94-2.amzn2023.0.2	amazonlinux	273 k
fonts-filesystem	noarch	1:2.0.5-12.amzn2023.0.2	amazonlinux	9.5 k
freetype	x86_64	2.13.2-5.amzn2023.0.2	amazonlinux	422 k
glib	x86_64	5.2.1-9.amzn2023.0.4	amazonlinux	49 k
google-noto-fonts-common	noarch	20240401-1.amzn2023.0.2	amazonlinux	17 k
google-noto-sans-vf-fonts	noarch	20240401-1.amzn2023.0.2	amazonlinux	593 k
graphite2	x86_64	1.3.14-7.amzn2023.0.3	amazonlinux	97 k
harfbuzz	x86_64	7.0.0-2.amzn2023.0.2	amazonlinux	873 k
java-17-amazon-corretto-headless	x86_64	1:17.0.20+10-1.amzn2023.1	amazonlinux	91 M
javapackages-filesystem	noarch	6.0.0-7.amzn2023.0.6	amazonlinux	12 k
javapackages-core-font-en	noarch	3.0-21.amzn2023.0.4	amazonlinux	10 k
libICE	x86_64	1.1.1-3.amzn2023.0.1	amazonlinux	76 k
libXt	x86_64	1.2.4-3.amzn2023.0.1	amazonlinux	45 k
libXau	x86_64	1.4.10-2.amzn2023.0.1	amazonlinux	639 k
libXdmcp	x86_64	1.4.10-2.amzn2023.0.1	amazonlinux	147 k
libXext	x86_64	1.3.6-1.amzn2023.0.1	amazonlinux	33 k
libXft	x86_64	1.4.2-1.amzn2023.0.1	amazonlinux	42 k
libXinerama	x86_64	1.1.5-6.amzn2023.0.1	amazonlinux	16 k
libXrandr	x86_64	1.5.4-3.amzn2023.0.1	amazonlinux	29 k
libXrender	x86_64	0.9.11-6.amzn2023.0.1	amazonlinux	79 k
libXtst	x86_64	1.3.0-3.amzn2023.0.1	amazonlinux	183 k
libXtst	x86_64	1.2.5-1.amzn2023.0.1	amazonlinux	22 k
libxtst	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k

519 kB/s | 69 kB 00:00

- Create a bash script to check if the Nginx service is running, if not running then script should start the service. Create vi check nginx.bash

```
#!/bin/bash

if systemctl is-active --quiet nginx
then
    echo "Nginx is running"
else
    echo "Nginx is not running"

    echo "nginx started"
fi
```

```
root@ip-172-31-32-133 ~]# vi checknginx.bash
root@ip-172-31-32-133 ~]# chmod 755 checknginx.bash
root@ip-172-31-32-133 ~]# ./checknginx.bash
nginx is not running
nginx started
root@ip-172-31-32-133 ~]# |
```

7. Create a bash script for a calculator. Create vi calculator.bash

```
#!/bin/bash

echo "Enter number"
read a

echo "Enter number"
read b

echo "addition = $((a+b))"

echo "subtraction = $((a-b))"

echo "Mulplication = $((a+b))"

echo "division = $((a+b))"
```

```
[root@ip-172-31-47-122 ~]# vi calculator.bash
[root@ip-172-31-47-122 ~]# chmod 755 calculator.bash
[root@ip-172-31-47-122 ~]# ./calculator.bash
Enter number
10
Enter number
20
addition = 30
subtraction = -10
Mulplication = 30
division = 30
```

8. Create a bash script to check if a directory exists, if not then create a directory. Create vi create_dir.bash

```
#!/bin/bash

if [ -d "$1" ]
then
    echo "Directory already exist"
else
    mkdir "$1"
    echo "Directory created sucessfully"
fi

~
```

```
[root@ip-172-31-47-122 ~]# vi createdir.bash
[root@ip-172-31-47-122 ~]# chmod 755 createdir.bash
[root@ip-172-31-47-122 ~]# ./createdir.bash
mkdir: cannot create directory '': No such file or directory
Directory created sucessfully
[root@ip-172-31-47-122 ~]#
```

9. Create a bash script to delete the last 3 lines of a file. Create vi test.txt

```
[root@ip-172-31-47-122 ~]# vi test.tx
[root@ip-172-31-47-122 ~]# chmod 755 test.txt
[root@ip-172-31-47-122 ~]# ./test.txt
Last 3 lines deleted successfully
[root@ip-172-31-47-122 ~]#
```

```
[root@ip-172-31-47-122 ~]# vi test.txt
[root@ip-172-31-47-122 ~]# ./delete3.bash test.txt
last 3 lines deleted
[root@ip-172-31-47-122 ~]# cat test.txt
line 1
line 2
line 3
[root@ip-172-31-47-122 ~]# |
```

10. Bash script to monitor cpu and if it is more than 80% then send email notification. Create vi cpu.bash

```
#!/bin/bash

cpu=$(top -bn1 | grep "Cpu(s)" | awk '{print 100 - $')

echo "CPU Usage: $cpu%"
if [ $cpu%.*} - gt 80 ]
then
    echo "Cpu usage above 80%"
else
    echo "Cpu usage is normal"
fi
```

```
cpu usage is normal
[root@ip-172-31-47-122 ~]# vi cpu.bash
[root@ip-172-31-47-122 ~]# chmod 755 cpu.bash
[root@ip-172-31-47-122 ~]# ./cpu bash
-bash: ./cpu: No such file or directory
[root@ip-172-31-47-122 ~]# ./cpu.bash
CPU Usage: 3.2%
./cpu.bash: line 6: [: syntax error: `-' unexpected
Cpu usage is normal
[root@ip-172-31-47-122 ~]# |
```

11. Bash script to monitor disk space and if it is more than 80% then send email notification.
Create disk.bash

```
#!/bin/bash

disk=$(df / | awk 'NR==2 {print $5}' | tr -d '%')

echo "Disk Usage: $disk%"

if [ $disk -gt 80 ]
then
    echo "Disk usage is above 80%"
else
    echo "Disk usage is normal"
fi
```

```
[root@ip-172-31-47-122 ~]# vi disk space.bash
2 files to edit
[root@ip-172-31-47-122 ~]# chmod 755 disk space.bash
chmod: cannot access 'space.bash': No such file or directory
[root@ip-172-31-47-122 ~]# ./disk space.bash
Disk Usage: 21%
Disk usage is normal
[root@ip-172-31-47-122 ~]# vi disk space.bash
```

12. Bash script to monitor memory and if it is more than 80% then send email notification.
Create memory.bash

```
#!/bin/bash

memory=$(free | awk 'Mem:/ {printf "%.0f", $3/$2 * 100}')

echo "Memory Usage: $memory%"

if [ $memory -gt 80 ]
then
echo "memory usage is above 80%"
else
echo "memory usage is normal"
fi
```

```
[root@ip-172-31-47-122 ~]# vi memory.bash
[root@ip-172-31-47-122 ~]# vi memory.bash
[root@ip-172-31-47-122 ~]# chmod 755 memory.bash
[root@ip-172-31-47-122 ~]# ./memory.bash
Memory Usage: 22f%
./memory.bash: line 7: [: 22f: integer expression expected
memory usage is normal
[root@ip-172-31-47-122 ~]#
```